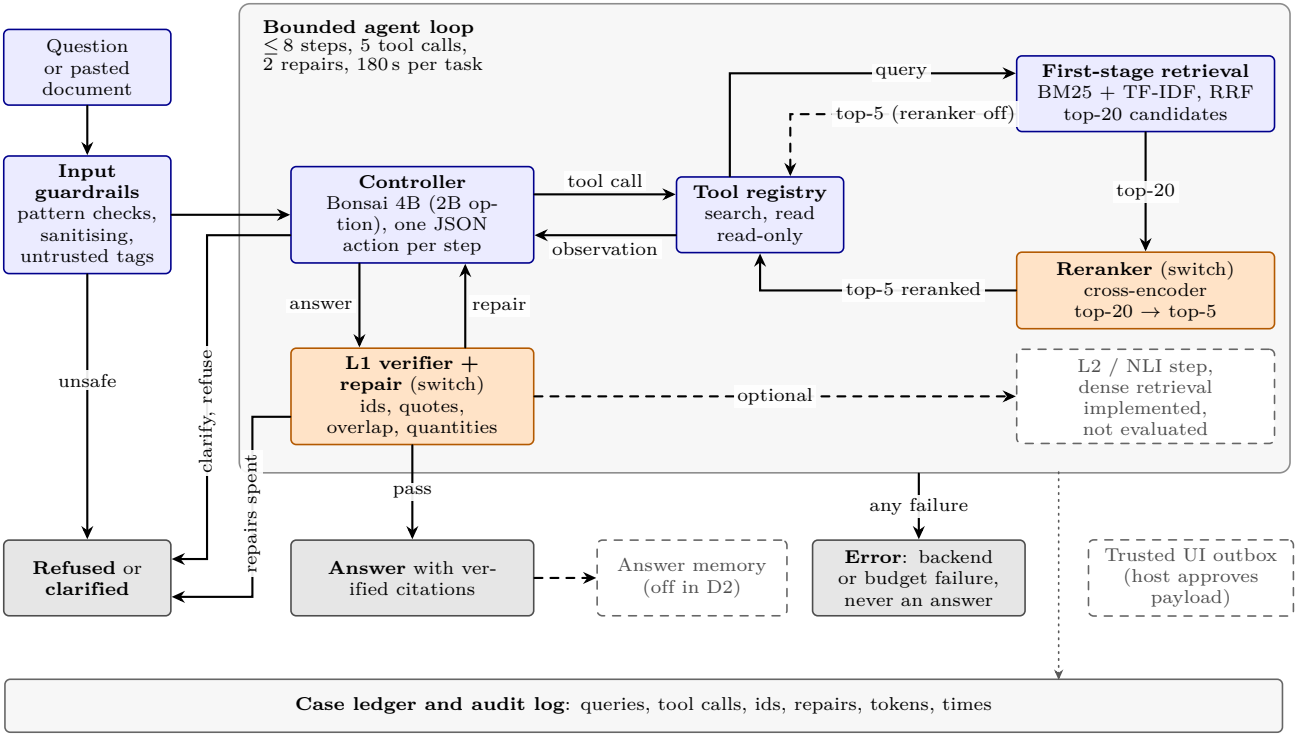

Bounded agent loop
≤ 8 steps, 5 tool calls,
2 repairs, 180 s per task
Question or pasted document
Input guardrails pattern checks, sanitising, untrusted tags
Controller Bonsai 4B (2B option), one JSON action per step
tool call
observation
Tool registry search, read read-only
query
top-5 (reranker off)
First-stage retrieval BM25 + TF-IDF, RRF top-20 candidates
top-20
Reranker (switch) cross-encoder top-20 → top-5
top-5 reranked
answer
repair
L1 verifier + repair (switch) ids, quotes, overlap, quantities
optional
L2 / NLI step, dense retrieval implemented, not evaluated
unsafe
clarify, refuse
repairs spent
pass
any failure
Refused or clarified
Answer with verified citations
Answer memory (off in D2)
Error: backend or budget failure, never an answer
Trusted UI outbox (host approves payload)
Case ledger and audit log: queries, tool calls, ids, repairs, tokens, times